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V-29—Legislative Rulings and Interpretations

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1.0 Purpose

The purpose of this bulletin is to provide uniform interpretation of particular sections of the Weights and Measures legislation.

2.0 Scope

This bulletin is applicable to all volumetric devices used in trade.

3.0 Definitions

All words and expressions used here have the same meaning as in the *Weights and Measures Regulations*.

4.0 Background

4.1 Some of the following information was previously set out in "Rulings and Interpretations" documents, which served to provide guidance for specific sections of the Weights and Measures legislation. The "Rulings and Interpretations" were internal documents typically consulted by Measurement Canada (MC) staff only. In light of alternative service delivery mechanisms such as accreditation and registration, it is critical that all

relevant information regarding the application of legislation be made available to all parties involved with their administration. For this reason, information that remains relevant has been transferred to this bulletin.

5.0 Policy

5.1 Multiple Deliveries at a Single Site (section 28 of the Act)

5.1.1 Under section 28 of the *Weights and Measures Act*, a person is guilty of an offence if the person allows a ticket to remain in a device at any time other than the time at which the quantity of the commodity is being measured.

5.1.2 With regard to vehicle-mounted liquid meters, a delivery is sometimes made to multiple tanks for a single customer at one site. In this case, the ticket may remain in the device between the delivery to each tank, provided that the vehicle is not in motion on a public street, highway, or thoroughfare between deliveries and the meter ticket contains the statement "Multiple Deliveries at One Site" or the French equivalent (**Multiples livraison à un même site**) or words that have the same meaning.

5.2 Upgrading or Reinstallation of Software (Regulations, section 35.1)

5.2.1 Section 35.1 of the Regulations requires anyone who alters a device in a way that might affect its accuracy, to also ensure that the device is calibrated immediately after the alteration to ensure that the measurement error is as close to zero as possible.

5.2.2 Given the near infinite range of possible software changes and the unexpected or unknown consequences that such changes might have on a device's performance, MC has deemed the installation of upgraded software as an alteration which might affect the accuracy of a device. Therefore, any time software is upgraded or reinstalled in a previously inspected device, the installer must ensure that the device is calibrated prior to being put back into service, and that any residual errors are less than 1/3 of the applicable limit of error.

5.2.3 Simply updating calibration factors and/or parameter settings in the newly installed software with those employed by the previous software version is not considered adequate to meet the requirements of subsection 35(1) of the Regulations.

5.3 Positive Displacement Meter Factors Derived From Other Specific Liquids (Regulations, section 237)

5.3.1 Section 237 of the Regulations allows meters to be tested and calibrated using a liquid other than the liquid that the meter will measure in trade, provided the test liquid has similar hydraulic properties.

5.3.2 Subsection 5.3 of this bulletin only applies to positive displacement (PD) meter types that rely on the liquids being measured to mechanically displace components in the meter. It does not apply to turbine or Coriolis (mass flow) meters.

5.3.3 Prior to the advent of electronic registers that allow for multiple meter factors (MFs) for measuring multiple products, it was MC's policy to calibrate, with a single MF, PD meters that measure both gasoline and diesel on the most viscous product (diesel). In these cases, the meter would over-deliver on the less viscous product and ensure that the customer would get at least the correct amount of product offered for sale.

5.3.4 In situations where certain products are not typically available on a year-round basis or difficult to obtain for the inspection of a PD meter, it is desirable to use alternate liquids as allowed by section 237 of the Regulations. In addition, because using the actual product to inspect a meter renders it unusable, it is desirable to use alternate liquids. The following table describes the circumstances where alternate liquids may be used for inspection purposes and stipulates which liquids may be used as an alternate.

Table 1

Circumstance	Liquid to be measured	Alternate liquid allowed for inspection purposes
Vehicle-mounted metering system that normally delivers gasoline and diesel, but on a seasonal or occasional basis delivers methanol. No methanol is stored on the site.	Methanol	Diesel fuel
Vehicle-mounted metering system that normally delivers gasoline and diesel, but on an occasional basis delivers Avgas (Avgas cannot be returned to storage if it has been measured through a non-dedicated metering system).	Avgas	Gasoline

Circumstance	Liquid to be measured	Alternate liquid allowed for inspection purposes
Vehicle-mounted metering system that normally delivers gasoline and diesel, but on an occasional basis delivers Jet A1 fuel (Jet A1 fuel cannot be returned to storage if it has been measured through a non-dedicated metering system).	Jet A1 fuel	Diesel fuel
Any metering system that is dedicated to delivering anhydrous ammonia (NH_3) where tests are performed at the dealer's premises on a bench and under controlled conditions (due to NH_3 safety requirements for storage and handling and toxicity hazards).	Anhydrous ammonia (NH_3)	Propane

5.3.5 In cases where an alternative liquid is used for inspection purposes, all other relevant information on the actual product being sold must be correct. This includes, but is not limited to, the product name, product density and the tables used to perform temperature conversion. For further information on the product being sold, refer to Measurement Canada Bulletin V-18 — Selection of Volume Correction Factor Tables and Standard Density Values for some Common Products.

5.3.6 Alternate liquids cannot be used for inspection purposes for the sake of convenience.

5.3.7 In cases where an alternate liquid is used for inspection purposes, the certificate must be annotated with the following statement or a similar statement which has the same meaning:

Meter # "X" was not tested on [product name]. Pulses per litre of XXX.XXX from [alternate liquid name] used. This may result in under-registration errors for this product.

Traders should be made aware of this statement prior to having them sign the certificate.

5.3.8 The above table may only be revised where such revisions are supported with relevant data.

5.4 Interlock Mechanisms for Propane Dispensers (Regulations, section 254)

5.4.1 Meters referred to in section 254 of the Regulations must have an interlock mechanism which prevents their use unless the register has been reset to zero.

5.4.2 In the case of propane dispensers, the "holster switch" is the method most often used. Activation of this switch is necessary before the nozzle can be returned to its standby position. The dispenser is not operable until the registering elements are at zero.

5.4.3 A second acceptable method is the "dead-man switch". The operator must manually hold the switch to use the dispenser. If the switch is released, the dispenser stops. The registering elements must be reset to zero before the next delivery.

5.4.4 Some electronic registers have an optional "time out feature" that shuts the dispenser off if the register does not receive any pulse signals within a given time period (i.e. 15, 30, 45 seconds). The registering

elements must therefore be reset to zero before the dispenser can be used again. For propane dispensers only, this method is also acceptable provided that the setting of the time delay does not exceed 45 seconds.

5.5 Volume Corrected to 15 °C (Regulations, section 295)

5.5.1 Except for retail applications (refer to Bulletin V-20 — Printer Requirements for Volumetric Liquid Meters Equipped with Automatic Temperature Compensation) where the primary register is marked, any liquid measuring device equipped with an automatic temperature compensator, or other compensator, and a printer, must issue tickets bearing a statement that the volume has been adjusted to a standard condition stated on the ticket. This statement must be in close proximity to the volume declaration. With regard to temperature compensation, the following are examples of permitted statements:

Volume corrected to 15 °C / *Volume corrigé à 15 °C*
Vol. corrected to 15 °C / *Vol. corrigé à 15 °C*
Corrected litres at 15 °C / *Litres corrigés à 15 °C*
Vol. correction to 15 °C / *Correction du vol. à 15 °C*

5.5.2 The word "corrected" (*corrigé*) may be replaced by "adjusted" (*ramené*). The word "at" (*à*) is interchangeable with "@". The word "degrees" (*degrés*) or abbreviation "deg." is interchangeable with the symbol "°".

5.5.3 The statement must contain the words "correction", "corrected", or "adjusted" to indicate that volume has, in fact, been changed.

5.5.4 It should be noted that under section 21 of the Regulations, a volumetric liquid meter equipped with an automatic temperature compensator must be marked with the words "Volume corrected to 15 °C"

or "*Volume corrigé à 15 °C*" immediately adjacent to the registration of net quantity on the device. Other words and/or abbreviations are not acceptable.

5.6 Approval and Inspection of Gross Volume Indications on Registers at Loading Racks (SVM-2, subsection 5(1))

5.6.1 Bulk fuel is normally picked up and delivered by specialized trucking companies or contract drivers. The driver is typically not a party to the actual transaction, but rather, just a carrier. The driver's primary concern is to not overload the truck and cause a spill. For this reason, most registers used on loading racks only display the gross volume delivered, as this is the actual volume going into the truck and the value which is important in the prevention of overloads. In fact, the industry believes that if both the net and gross were displayed, it could lead to confusion and actually cause the overload. It should be noted that these transactions are all accompanied with a bill of lading which has all the pertinent information on them.

5.6.2 Registers destined for use on loading rack applications must be capable of displaying the net volume, but may be configured in the field to display both net and gross volumes or gross volume alone. In all cases, the unit displayed must be clearly identified. This is based on the current industry practice whereby indications at the loading rack are normally displayed in gross volume only. This interpretation of SVM-2, subsection 5(1) does not apply to devices used at the retail level of trade or where the persons present at the transaction have an interest in the actual unit of trade measurement.

5.7 Interpretation of "on Demand for Inspection Purposes" (SVM-2, subsection 5(1) and section 8)

5.7.1 Subsection 5(1) and section 8 of SVM-2 require a register incorporating automatic temperature compensation to be designed to register the net volume and, on demand for inspection purposes, the gross volume and the temperature of the metered liquid.

5.7.1.1 In this subsection and section, the phrase "on demand for inspection purposes" means that the gross volume and temperature indications must be available in real time during a test run of the system. In order to meet the intent of this definition, it is not acceptable to only provide the gross volume and average temperature over time as a printed or indicated record of values after a test run has been completed.

5.7.1.2 Any new or modified electronic register designs to be approved in Canada will be required to comply with this interpretation and provide acceptable means to view in real time the indications of gross volume and temperature during a test run of the system.

5.7.2 Live indications of gross volume and temperature may be provided on a supplementary display at all times or at any time required during an inspection test run of the system.

5.7.2.1 Means to activate indications of gross volume and temperature registrations shall be readily accessible and permanently installed without having to disassemble any part of the register's housing, which has been sealed to protect metrological adjustment.

5.7.3 The use of a key switch, remote control device (infrared or otherwise), magnetic switch, removable key pad or specialised tools is not permitted.

5.8 Use of Secondary Means of Indication (Regulations, section 143)

5.8.1 Section 143 allows for the installation of a secondary means of indication where the primary means of indication cannot be clearly seen.

5.8.2 Secondary means of indication means a display used in conjunction with a primary mechanical register or approved electronic register that does not have means of adjustment that may affect the accuracy of the measuring system. Secondary means of indication do not require approval since they are not devices according to section 2 of the *Weights and Measures Act*.

5.8.3 Secondary means of indication must agree with one of the values stated in section 139 of the Regulations and must comply with section149 of the Regulations with respect to conformance to specifications for registration set out in Part V of the Regulations and Ministerial Specifications. Mechanical secondary indicators must meet the requirements regarding the minimum increment of registration, as set out in Bulletin V-07 — Increment of Registration for Volumetric Liquid Meters.

6.0 Revision

The original bulletin was issued on August 25, 2010 and became effective on October 1st, 2010.

6.1 The purpose of revision 1 is to add two new sections. Section 5.7 provides an interpretation of the phrase "on demand for inspection purposes" (SVM-2, subsection 5(1) and section 8), and section 5.8 allows for the use of non-approved secondary means of indication. Minor editorial corrections were also made.

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